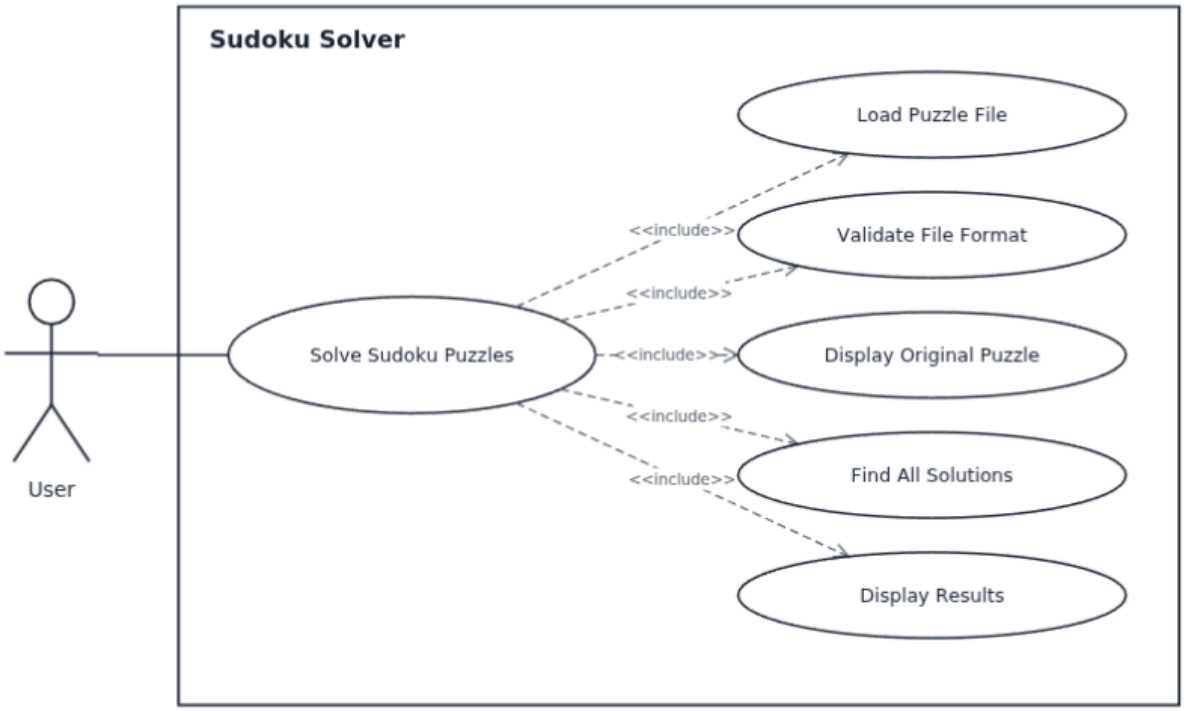
Use Case Diagram



1. Identify Use Case

Use case: Solve Sudoku Puzzles
System: Sudoku Solver
Actor: User
Goal: Obtain every solution for the five supplied puzzle files, or learn that a puzzle has no solution.

2. Brief Description

The user runs the Sudoku Solver to process puzzle1.txt through puzzle5.txt. For each readable, correctly formatted puzzle, the system displays its filename and original grid, then displays every valid solution or "No solution found". If a file cannot be loaded, the system reports the problem and continues to the next file.

3. Step-by-step flow of events

1. The user places puzzle1.txt through puzzle5.txt in the program's working directory and starts the program.
2. The system selects the next puzzle file and displays its filename.
3. The system opens the file and checks that it contains exactly 81 allowed cells, with whitespace permitted.
4. The system displays the original puzzle, using underscores for empty cells.
5. The system checks the starting digits for row, column, and block conflicts.

6. The system searches for every completion that preserves the starting digits and satisfies the Sudoku rules.
7. The system displays each solution when it is found. After searching, it displays the solution total, or "No solution found" when there are none.
8. The system repeats the file-processing steps until all five files have been attempted.
9. The user reviews the displayed puzzles, solutions, and any messages.

4. Detailed Description

The user places the five puzzle files in the working directory and starts the program. The system processes puzzle1.txt through puzzle5.txt in order.

For each file, the system displays its filename and checks that it contains exactly 81 cells using digits 1 through 9 or underscores. Whitespace is allowed. If loading fails, the system explains the error and continues to the next file.

After loading a puzzle, the system displays the original grid and checks the starting digits for conflicts. It then finds all solutions that preserve those digits and follow the row, column, and 3x3 block rules.

Each solution is displayed when found, followed by the total after searching finishes. If the starting digits conflict or no solution exists, the system displays “No solution found.” Processing continues until all five files have been attempted. The original files remain unchanged.

GenAI Assistance: ChatGPT was used for assistance with making and analyzing the Use Case Diagram